

Cloud LearnX

Mathematics

Mixed Algebra and Geometry — Practice Worksheet

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Topic: Mixed Algebra and Geometry

Instructions:

- Answer ALL questions.
- Show ALL working — method marks are awarded for correct steps.
- Write answers in the spaces provided.

Answer ALL ALL questions.
Write your answers in the spaces provided.
You must write down all the stages in your working.

1 Simplify fully:

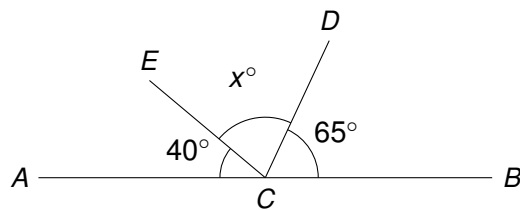
$$5(2x + 3) - 4x$$

.....

(2)

(Total for Question 1 is 2 marks)

2 The diagram shows three straight lines meeting at point C . A , C and B lie on a straight line.



Work out the size of angle x .

$$x = \dots\dots\dots^\circ$$

(2)

(Total for Question 2 is 2 marks)

3 Solve the equation.

$$5x + 7 = 32$$

$$x = \dots\dots\dots$$

(2)

(Total for Question 3 is 2 marks)

4 A rectangular science lab bench has a length of 120 cm and a width of 60 cm.



Work out the perimeter of the bench.

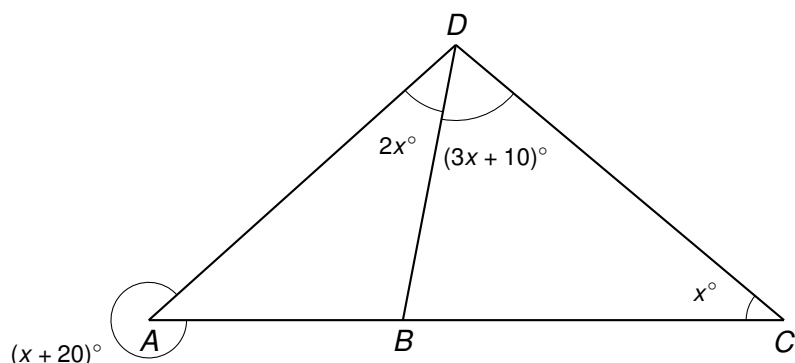
..... cm

(2)

(Total for Question 4 is 2 marks)

5 ABC is a straight line. D is a point above the line, with rays drawn from D to points A , E , C where E lies on AC between A and C .

The three angles formed at point on the line by the rays from D are algebraic expressions, as shown in the diagram.



In triangle ABD , the angle at A is $(x + 20)^\circ$ and angle $ADB = 2x^\circ$.

In triangle CBD , the angle at C is x° and angle $BDC = (3x + 10)^\circ$.

Angle ABD and angle DBC lie on the straight line ABC .

Since angles ABD and DBC are angles on a straight line, and using the fact that angles in each triangle sum to 180° , it can be shown that:

$$2x + (3x + 10) = 180$$

Use this equation to work out the value of x .

$$x = \dots\dots\dots$$

(3)

(Total for Question 5 is 3 marks)

6 Expand and simplify $(x + 3)(x - 5)$.

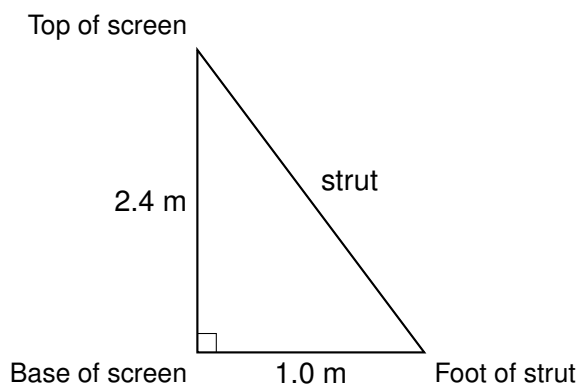
.....

(3)

(Total for Question 6 is 3 marks)

7 As part of a Model United Nations exhibition, a diagonal support strut is fitted to hold up a large display screen. The strut, the vertical screen and the horizontal base form a right-angled triangle, as shown in the diagram.

The screen stands vertically and is 2.4 m tall. The foot of the strut is fixed to the ground 1.0 m away from the base of the screen. The strut connects the top of the screen to this point on the ground.



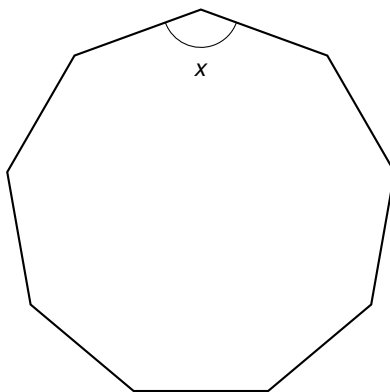
Calculate the length of the strut. Give your answer correct to 3 significant figures.

..... m

(3)

(Total for Question 7 is 3 marks)

8 The diagram shows a regular nonagon (a 9-sided polygon).



Regular nonagon (9 equal sides, 9 equal angles)

The diagram shows a regular nonagon (9 sides).

Work out the size of the interior angle marked x .

$x = \dots\dots\dots^\circ$
(3)

(Total for Question 8 is 3 marks)

- 9 A UI/UX design team is timing how quickly a new app screen loads for users testing a prototype. They use the formula for average speed of data transfer:

$$v = \frac{d}{t}$$

where v is the average speed (in megabytes per second, MB/s), d is the amount of data transferred (in megabytes), and t is the time taken (in seconds).

During one test, a badminton club's booking app transfers $d = 45$ megabytes of data in $t = 7.5$ seconds.

$$v = \dots\dots\dots \text{MB/s}$$

- (a) Work out the average transfer speed, v , during this test.

(2)

- (b) In a second test, the app must transfer 60 megabytes of data. For a smooth user experience, the average transfer speed must be at least 8 MB/s.

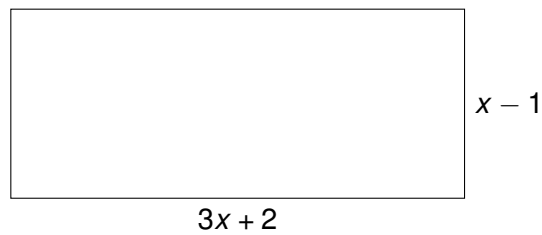
Work out the maximum time, in seconds, that this transfer can take.

$$t = \dots\dots\dots \text{s}$$

(1)

(Total for Question 9 is 3 marks)

- 10** A rectangular badminton court marking is being designed for a UI/UX mockup. The rectangle has a length of $(3x + 2)$ cm and a width of $(x - 1)$ cm.



The perimeter of the rectangle is 42 cm.

- (a) Show that $8x + 2 = 42$, and hence find the value of x .

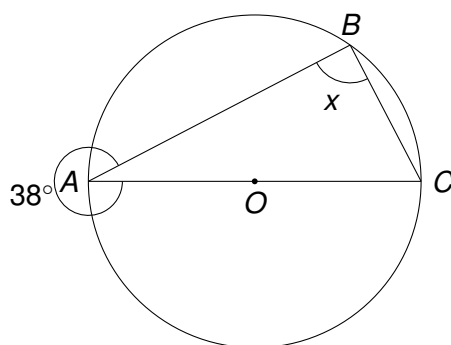
$x = \dots\dots\dots$
(2)

- (b) Work out the area of the rectangle.

$\dots\dots\dots \text{ cm}^2$
(2)

(Total for Question 10 is 4 marks)

11 A , B and C are points on the circumference of a circle with centre O . AC is a diameter of the circle.



Angle $BAC = 38^\circ$.

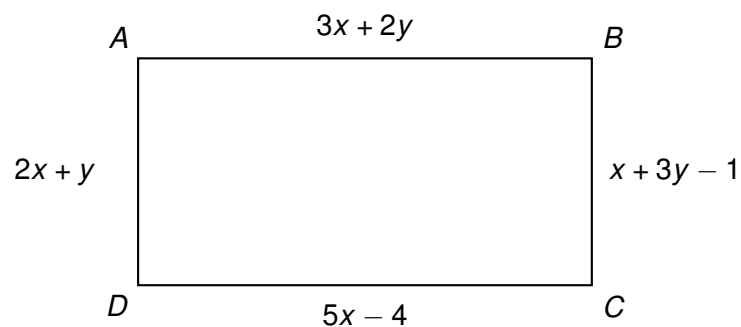
Work out the size of angle x , giving a reason for your answer.

$x = \dots\dots\dots^\circ$
(3)

(Total for Question 11 is 3 marks)

12 A UI/UX designer is laying out a rectangular badminton court icon for a sports app. The court-shaped rectangle $ABCD$ has:

- top side $AB = 3x + 2y$ cm
- bottom side $DC = 5x - 4$ cm
- left side $AD = 2x + y$ cm
- right side $BC = x + 3y - 1$ cm



Since $ABCD$ is a rectangle, opposite sides are equal in length.

(a) Use this fact to write down two equations connecting x and y .

.....

.....

(2)

(b) Solve your equations simultaneously to find the value of x and the value of y .

$$x = \dots\dots\dots$$

(1)

$$y = \dots\dots\dots$$

(1)

(Total for Question 12 is 4 marks)

13 A rectangular badminton court design in a UI/UX mockup has its length and width given by algebraic expressions.

The rectangle has width $(2x - 3)$ cm and length $(x + 5)$ cm.

A designer removes a smaller rectangular icon of width $(x - 1)$ cm and length $(x + 2)$ cm from the layout.

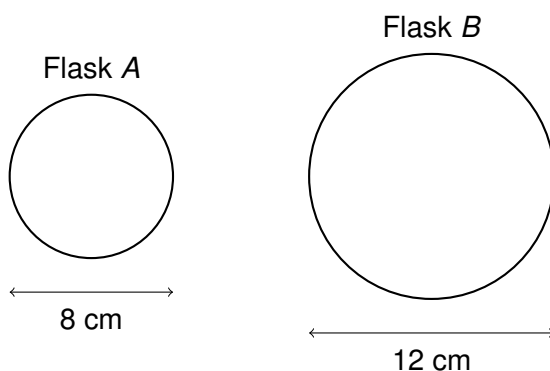
Show that the remaining area, $A \text{ cm}^2$, is given by

$$A = x^2 + 8x - 17$$

(3)

(Total for Question 13 is 3 marks)

- 14** In a school science lab, two spherical flasks used for a chemistry demonstration are mathematically similar in shape. Flask *A* has a diameter of 8 cm and flask *B* has a diameter of 12 cm.



The curved surface of flask *A* has an area of 201 cm^2 .

- (a) Find the linear scale factor from flask *A* to flask *B*.

.....
(1)

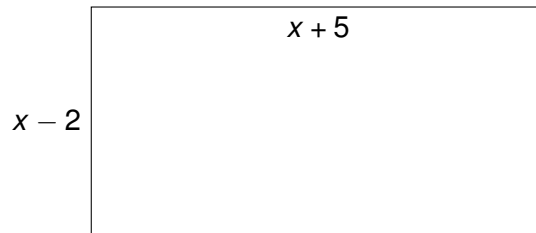
- (b) Work out the curved surface area of flask *B*.

Give your answer to 3 significant figures.

..... cm^2
(3)

(Total for Question 14 is 4 marks)

15 A laboratory petri dish is rectangular in shape. The length is $(x + 5)$ cm and the width is $(x - 2)$ cm.



The area of the petri dish is 54 cm^2 .

(a) Show that $x^2 + 3x - 64 = 0$. **(2)**

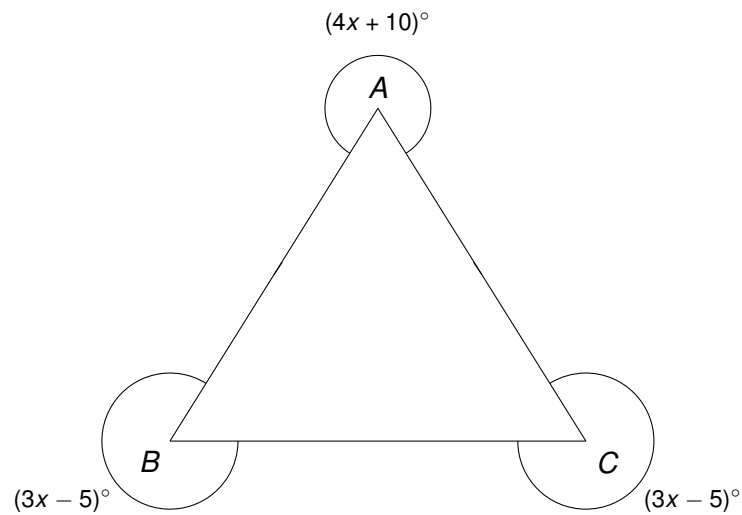
(b) Find the width of the petri dish, giving your answer correct to 2 decimal places. cm
(2)

(Total for Question 15 is 4 marks)

16 Triangle ABC is isosceles, with $AB = AC$.

The angles of the triangle are given, in degrees, by:

$$\angle BAC = (4x + 10)^\circ, \quad \angle ABC = \angle ACB = (3x - 5)^\circ$$



Show that $x = 17$, and hence show that triangle ABC must in fact be equilateral.

(4)

(Total for Question 16 is 4 marks)

TOTAL FOR PAPER IS 49 MARKS